GAME BASED APTITUDE

60 PYQ-Style Questions • Practice Set 2

Placement & Competitive Exam Practice Series

Included	60 PYQ-Style Questions • Placement-focused
Useful For	TCS NQT • Cognizant • Infosys • Wipro • Accenture • Campus Placements
Topics Covered	Grid Challenge • Digit Challenge • Motion Challenge • Geo-Symmetry • Switch Challenge • Inductive Patterns
Format	10 MCQs + Answer Key per Page • Practice Set 2

These are PYQ-style practice questions modeled on common campus recruitment exam patterns.
They are designed for comprehensive practice, mental speed, and spatial logic assessment.

GAME BASED APTITUDE • GRID CHALLENGE & VISUAL MEMORY

Practice Set 2 • Section 1 of 6 • 10 Questions with Answer Key

Q1. In a 5x5 memory grid, 7 green markers illuminate for 2 seconds. What memory mechanism retains visual icons under 500ms?

A) Iconic Sensory Memory
B) Long-term procedural
C) Semantic episodic
D) Auditory rehearsal

Q2. A token is placed at (2, 2) in a 4x4 matrix. It moves 1 step Up, 2 steps Right. New position coordinates are:

A) (1, 4)
B) (3, 4)
C) (1, 2)
D) (2, 4)

Q3. In an spatial recall challenge, the entire board is mirrored vertically (flipped top-to-bottom). Cell (1, 1) in 4x4 moves to:

A) (4, 1)
B) (1, 4)
C) (4, 4)
D) (2, 1)

Q4. How many total 1x1 squares exist in a standard 6x6 grid?

A) 36
B) 91
C) 49
D) 30

Q5. In a path memory challenge, a snake sequence visits 8 consecutive cells. If you repeat in reverse order, which is first?

A) The 8th visited cell
B) The 1st visited cell
C) The center cell
D) Random cell

Q6. A 3x3 grid has 9 colored lights. If 4 lights are Red and remaining are Yellow, probability of picking Yellow at random is:

A) 5/9
B) 4/9
C) 1/2
D) 2/3

Q7. In a spotlight scanning game, peripheral vision detects which visual stimuli quickest?

A) Rapid motion and flicker
B) Fine text details
C) Hue saturation differences
D) Roman numerals

Q8. Two identical shapes are placed on an 8x8 checkerboard. Minimum distance between two diagonal corners of a 1x1 square is:

A) sqrt(2) units
B) 1 unit
C) 2 units
D) 1.5 units

Q9. In a maze grid without obstacles, shortest Manhattan distance between (1, 1) and (5, 4) is:

A) 7 steps
B) 5 steps
C) 9 steps
D) 6 steps

Q10. When a 4x4 grid rotates 180 degrees, which cell position maps onto itself?

A) None (even dimension has no center cell)
B) (2, 2)
C) (3, 3)
D) (1, 1)

ANSWER KEY & SOLUTIONS — GRID CHALLENGE & VISUAL MEMORY (SET 2)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • DIGIT CHALLENGE & MENTAL ARITHMETIC

Practice Set 2 • Section 2 of 6 • 10 Questions with Answer Key

Q1. Using numbers [3, 4, 8], how can target 32 be formed using basic operations?

A) 4 * 8
B) (8 + 3) * 4
C) 3 * 8 + 4
D) 8 * 3 - 4

Q2. In a speed calculation round, evaluate: 25 * 16 without paper:

A) 400
B) 350
C) 420
D) 450

Q3. Missing number in series: 5 -> 11 -> 23 -> 47 -> ?

A) 95 (multiply by 2, add 1)
B) 94
C) 92
D) 99

Q4. Evaluate rapid mental sum: 88 + 47 + 12 + 53 = ?

A) 200
B) 190
C) 210
D) 205

Q5. If 3 apples and 2 oranges cost $18, and 1 apple costs $4, how much does 1 orange cost?

A) $3
B) $2.50
C) $4
D) $5

Q6. Rapid power challenge: 2 raised to power 7 equals:

A) 128
B) 64
C) 256
D) 512

Q7. If (x / 4) + 6 = 15, what is the value of x?

A) 36
B) 24
C) 40
D) 32

Q8. In a digit challenge target game, what is 15% of 240 calculated mentally?

A) 36
B) 32
C) 40
D) 28

Q9. Consecutive integers sum to 51: x + (x+1) + (x+2) = 51. The middle integer is:

A) 17
B) 16
C) 18
D) 15

Q10. Operation rule: a @ b = (a * b) - (a + b). What is 6 @ 4?

A) 14
B) 16
C) 12
D) 18

ANSWER KEY & SOLUTIONS — DIGIT CHALLENGE & MENTAL ARITHMETIC (SET 2)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • MOTION CHALLENGE & TRAJECTORY PHYSICS

Practice Set 2 • Section 3 of 6 • 10 Questions with Answer Key

Q1. In a speed-distance simulation, a vehicle traveling at 72 km/h converts to how many meters per second?

A) 20 m/s
B) 25 m/s
C) 15 m/s
D) 30 m/s

Q2. Two runners start at the same point on a 400m circular track in opposite directions at 4 m/s and 6 m/s. They meet in:

A) 40 seconds
B) 50 seconds
C) 60 seconds
D) 30 seconds

Q3. In a pulley simulator, a single fixed pulley provides a mechanical advantage of:

A) 1
B) 2
C) 3
D) 0.5

Q4. If an object moves in uniform circular motion at constant speed, its acceleration is directed:

A) Towards the center (Centripetal)
B) Tangentially forward
C) Outwards away from center
D) Zero

Q5. A boat crosses a 100m wide river flowing East at 3 m/s. Boat steers North at 4 m/s. Resultant velocity is:

A) 5 m/s
B) 7 m/s
C) 1 m/s
D) 6 m/s

Q6. In a ball trajectory game, launching a projectile at which angle achieves maximum horizontal range in vacuum?

A) 45 degrees
B) 30 degrees
C) 60 degrees
D) 90 degrees

Q7. An object dropped from rest falls freely under gravity (g=9.8 m/s^2). Its speed after 2 seconds is:

A) 19.6 m/s
B) 9.8 m/s
C) 4.9 m/s
D) 39.2 m/s

Q8. If Gear 1 (10 teeth) turns Gear 2 (30 teeth), what is the torque multiplication factor on Gear 2?

A) 3x increase
B) 3x decrease
C) 1x (same)
D) 9x increase

Q9. In an aerial avoidance round, an aircraft flying North turns 270 degrees clockwise. Its new heading is:

A) West
B) East
C) South
D) Northwest

Q10. When stepping off a small floating boat onto a dock, why does the boat drift backwards?

A) Newton third law of action-reaction
B) Water surface tension
C) Archimedes principle
D) Coriolis effect

ANSWER KEY & SOLUTIONS — MOTION CHALLENGE & TRAJECTORY PHYSICS (SET 2)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • GEO-SYMMETRY & SPATIAL FOLD REASONING

Practice Set 2 • Section 4 of 6 • 10 Questions with Answer Key

Q1. A circular paper disk is folded along its diameter twice (into a quarter circle). How many layers of paper are there?

A) 4 layers
B) 2 layers
C) 8 layers
D) 6 layers

Q2. How many total faces, edges, and vertices does a regular octahedron have?

A) 8 faces, 12 edges, 6 vertices
B) 6 faces, 12 edges, 8 vertices
C) 8 faces, 16 edges, 8 vertices
D) 12 faces, 18 edges, 8 vertices

Q3. If you view a 3D letter 'E' in a flat vertical mirror, how does it appear?

A) Horizontally reversed (backward 3-like)
B) Upside down
C) Unchanged
D) Rotated 90 degrees

Q4. What 3D shape is generated by rotating a right-angled triangle around one of its perpendicular legs?

A) Right circular cone
B) Cylinder
C) Sphere
D) Paraboloid

Q5. A standard six-sided die has opposite faces summing to:

A) 7
B) 6
C) 8
D) 9

Q6. How many lines of reflective symmetry does an isosceles trapezoid have?

A) 1 line of symmetry
B) 2 lines of symmetry
C) 0 lines of symmetry
D) 4 lines of symmetry

Q7. Looking at a cube from directly above (bird eye orthographic view), what 2D shape is visible?

A) Square
B) Hexagon
C) Triangle
D) Trapezoid

Q8. In a block counting spatial puzzle, a 3x3x3 cube is painted red on outside and cut into 1x1x1 unit cubes. How many have NO red paint?

A) 1 cube (the central core)
B) 8 cubes
C) 6 cubes
D) 0 cubes

Q9. How many unit cubes in the painted 3x3x3 cube have exactly 3 painted faces?

A) 8 (the corners)
B) 12
C) 6
D) 4

Q10. A regular polygon with interior angles of 120 degrees each is a:

A) Regular Hexagon
B) Regular Pentagon
C) Regular Octagon
D) Equilateral Triangle

ANSWER KEY & SOLUTIONS — GEO-SYMMETRY & SPATIAL FOLD REASONING (SET 2)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • SWITCH CHALLENGE & LOGIC GATE NETWORKS

Practice Set 2 • Section 5 of 6 • 10 Questions with Answer Key

Q1. A NOR gate gives a TRUE (1) output under which condition?

A) Only when both inputs are 0
B) When both inputs are 1
C) When at least one input is 1
D) Always 1

Q2. In a 3-way staircase lighting system, two switches control 1 lamp. Changing either switch status always:

A) Toggles the lamp on/off
B) Turns it permanently off
C) Blows the fuse
D) Has no effect

Q3. According to De Morgan laws of Boolean algebra, NOT(A AND B) is equivalent to:

A) (NOT A) OR (NOT B)
B) (NOT A) AND (NOT B)
C) A OR B
D) NOT(A OR B)

Q4. In a binary counter switch puzzle with 3 bits [B2, B1, B0], what is the next state after 011 (3)?

A) 100 (4)
B) 101 (5)
C) 010 (2)
D) 110 (6)

Q5. What is the boolean evaluation of: 1 AND (0 OR 1) AND (NOT 0)?

A) 1
B) 0
C) Undefined
D) High impedance

Q6. An XNOR gate output is 1 when:

A) Both inputs are identical (00 or 11)
B) Inputs are different (01 or 10)
C) Input A is 1 only
D) Input B is 1 only

Q7. How many distinct input combinations are represented in a truth table for 4 binary variables?

A) 16 combinations (2^4)
B) 8 combinations
C) 32 combinations
D) 12 combinations

Q8. In an electric relay circuit, an energized normally-open (NO) contact will:

A) Close and conduct current
B) Open and break current
C) Reverse polarity
D) Stay static

Q9. If switch A is ON, switch B is OFF, and circuit rule is (A OR B) AND (NOT B), circuit state is:

A) Active (1)
B) Inactive (0)
C) Oscillating
D) Shorted

Q10. What is the simplest equivalent gate for an inverter connected directly to the output of an OR gate?

A) NOR gate
B) NAND gate
C) XOR gate
D) AND gate

ANSWER KEY & SOLUTIONS — SWITCH CHALLENGE & LOGIC GATE NETWORKS (SET 2)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • INDUCTIVE REASONING & SEQUENCE PATTERNS

Practice Set 2 • Section 6 of 6 • 10 Questions with Answer Key

Q1. Next number in series: 3, 7, 15, 31, 63, ?

A) 127 (double and add 1)
B) 126
C) 125
D) 129

Q2. Pattern of geometric figures: 4 sides -> 8 sides -> 12 sides -> ?. Next figure has:

A) 16 sides
B) 14 sides
C) 20 sides
D) 15 sides

Q3. A symbol rotates 45 degrees counter-clockwise at each step. After 8 steps, it completes:

A) One full 360 degree rotation
B) Half rotation (180 deg)
C) Quarter rotation
D) Two rotations

Q4. Series: B, E, H, K, N, ?. What is the next letter?

A) Q (+3 alphabet step)
B) P
C) R
D) S

Q5. Number grid: [1, 4, 9], [16, 25, 36], [49, 64, ?]. What is the missing square number?

A) 81 (9 squared)
B) 100
C) 72
D) 90

Q6. Which shape does NOT belong with the others: Equilateral Triangle, Square, Regular Hexagon, Scalene Triangle?

A) Scalene Triangle (not regular/equilateral)
B) Square
C) Equilateral Triangle
D) Regular Hexagon

Q7. Sequence: 1, 1, 2, 3, 5, 8, 13, ?. What is the next Fibonacci number?

A) 21
B) 20
C) 19
D) 24

Q8. A sequence alternates: +2, *2, +2, *2. Starting at 1: 1 -> 3 -> 6 -> 8 -> 16 -> 18 -> ?

A) 36
B) 20
C) 32
D) 24

Q9. In a 3x3 symbol matrix, each row has 1 Circle, 1 Square, 1 Triangle. Each row also has [Red, Blue, Green]. Row 3 has Red Circle and Blue Square. Missing is:

A) Green Triangle
B) Blue Triangle
C) Red Square
D) Green Circle

Q10. Sequence of days: Monday, Wednesday, Saturday, Wednesday, ?. The pattern of skipping days (+1, +2, +3, +4) leads to:

A) Monday (+4 days from Wednesday)
B) Sunday
C) Tuesday
D) Friday

ANSWER KEY & SOLUTIONS — INDUCTIVE REASONING & SEQUENCE PATTERNS (SET 2)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A